

Predictive Maintenance

Anomaly Detection on the FEMTO-ST Bearing Dataset

Author:	Oumaima Yaakoubi
Program:	Industrial Maintenance and Instrumentation (IMI) INSAT, Tunisia
Date:	September 11, 2026
Dataset:	FEMTO-ST / IEEE PHM 2012 (via Hugging Face)
Tools:	Python, Google Colab, scikit-learn, pandas, Matplotlib
LinkedIn:	linkedin.com/in/oumaima-yaakoubi

Abstract

This report presents a complete predictive maintenance pipeline built entirely with free software and a public dataset. Using the FEMTO-ST bearing degradation dataset, time-domain features (RMS, kurtosis, peak-to-peak) and frequency-domain features (FFT band energies) are extracted from raw vibration signals. An Isolation Forest model is then trained to detect anomalies without labeled data. The results show that early fault signatures can be detected automatically, enabling condition-based maintenance decisions.

Contents

Abstract	1
1 Introduction: Why Bearings Matter	3
1.1 The Four Stages of Bearing Failure	3
2 Objective of This Project	3
3 The FEMTO-ST Dataset	4
3.1 Origin	4
3.2 Data Format	4
4 From Raw Signal to Features	4
4.1 Time-Domain Features	5
4.2 Frequency-Domain Features (FFT)	5
5 Methodology: Anomaly Detection with Isolation Forest	6
5.1 What is Anomaly Detection?	6
5.2 Why Isolation Forest?	6
5.3 Workflow	6
6 Results	7
6.1 Feature Evolution Over Time	7
6.2 FFT Band Energy	7
6.3 Anomaly Detection Results	8
7 Discussion	9
7.1 What Worked	9
7.2 Limitations	9
7.3 Lessons Learned	9

8 Conclusion	10
9 Tools and Environment	10
10 References	10

1 Introduction: Why Bearings Matter

A bearing is a mechanical component that supports a rotating shaft and reduces friction between moving parts. Bearings are found in almost every rotating machine: electric motors, pumps, turbines, helicopter rotors, conveyor belts and production line equipment. When a bearing fails, the machine stops. In an industrial setting, an unexpected stop means lost production, costly repairs and sometimes safety risks.

Key Insight

Bearings do not fail instantly. They degrade progressively, and the earliest signs are detectable in vibration data long before the machine stops.

1.1 The Four Stages of Bearing Failure

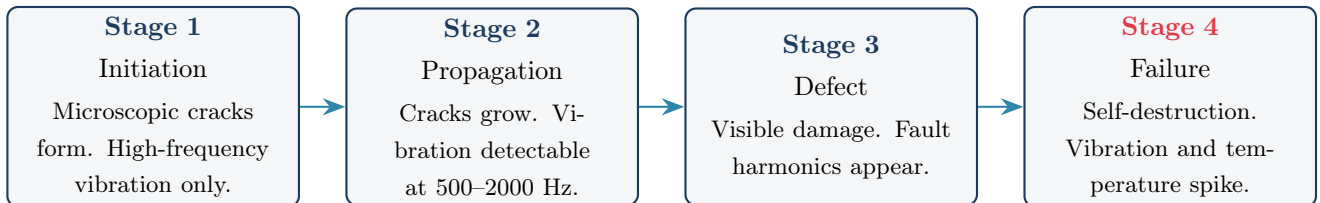


Figure 1: The four progressive stages of bearing degradation. Detection at the Stage 2 to Stage 3 transition enables preventive action.

If we can detect the transition from Stage 2 to Stage 3, we can schedule maintenance *before* the machine fails. This is the goal of **predictive maintenance**.

2 Objective of This Project

The objective is to build a **machine learning pipeline** that detects anomalies in bearing vibration data using **only software**, with no hardware required.

- Extract meaningful features from raw vibration signals using signal processing (time domain and frequency domain).
- Train an **unsupervised anomaly detection** model (Isolation Forest) on these features.
- Identify the moments when the bearing transitions from healthy to degrading.
- Document the full workflow as a reproducible portfolio project.

3 The FEMTO-ST Dataset

3.1 Origin

The FEMTO-ST dataset is a real-world dataset created by the French research institute **FEMTO-ST** for the **IEEE PHM 2012 Data Challenge**. Its purpose was to give researchers a benchmark for predicting the Remaining Useful Life (RUL) of bearings. It contains vibration recordings from **17 bearings** run until failure under controlled operating conditions.

3.2 Data Format

Each bearing was monitored continuously. Vibration was sampled at **25.6 kHz** in two axes: X (horizontal) and Y (vertical). The Hugging Face version `Amgharr/FEMTO-ST_DATASET` provides the raw acceleration signals.

Table 1: Columns in the raw dataset.

Column	Meaning
<code>ax</code>	Acceleration along X-axis
<code>ay</code>	Acceleration along Y-axis
<code>mag</code>	Magnitude (combined vibration)
<code>ax_diff</code> , <code>ay_diff</code> , <code>mag_diff</code>	First differences (rate of change)
<code>source_file</code>	Identifier of the original snapshot file

Each `source_file` (for example `acc_00001.csv`, `acc_00002.csv`) represents one vibration snapshot taken at a specific moment in the bearing's life. The files are ordered chronologically, so the row index represents time.

4 From Raw Signal to Features

The raw vibration signal contains tens of thousands of numbers per second. We must summarize each snapshot into a small set of meaningful numbers called **features**. This is the role of signal processing.

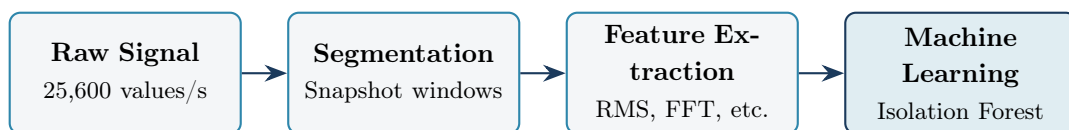


Figure 2: The transformation pipeline from raw vibration data to machine learning input.

4.1 Time-Domain Features

RMS (Root Mean Square)

Measures the overall energy or amplitude of the vibration:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

As a bearing degrades, it vibrates more violently, so RMS increases.

Kurtosis

Measures how spiky the signal is:

$$\text{Kurtosis} = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^4}{\sigma^4}$$

A perfect bell curve has kurtosis ≈ 3 . A bearing defect produces sharp spikes, so kurtosis rises above 3.

Peak-to-Peak

The range between maximum and minimum values:

$$\text{Peak-to-Peak} = \max(x) - \min(x)$$

4.2 Frequency-Domain Features (FFT)

The Fast Fourier Transform converts a signal from the time domain to the frequency domain. Instead of asking what the amplitude is at each moment, FFT answers how much energy is present at each frequency.

Table 2: Frequency bands used for FFT feature extraction.

Band	Frequency Range	Interpretation
Low	0–5 kHz	Shaft rotation energy
Mid	5–10 kHz	Fault harmonics
High	10–20 kHz	Early micro-crack signature

Leading Indicator

High-frequency energy appears *first*, when micro-cracks form, before RMS changes. This makes FFT band energy a leading indicator of degradation.

5 Methodology: Anomaly Detection with Isolation Forest

5.1 What is Anomaly Detection?

Anomaly detection is the process of automatically identifying data points that deviate significantly from the normal pattern. In our case, normal means healthy bearing vibration and anomaly means vibration that indicates degradation.

5.2 Why Isolation Forest?

Isolation Forest is an **unsupervised** algorithm. We do not need labeled data, that is, we do not need to know in advance which snapshots are faulty. The algorithm works by randomly partitioning the data:

- Normal points are surrounded by many similar points, so many random splits are needed to isolate them.
- Anomalies are few and different, so they are isolated quickly.

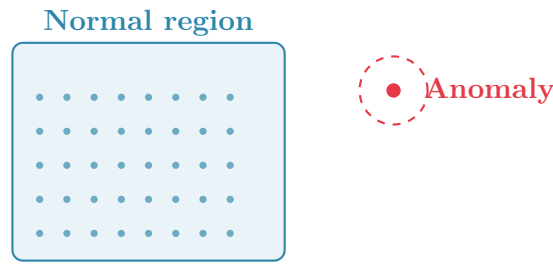


Figure 3: Conceptual illustration: anomalies are isolated by fewer random partitions than normal points.

5.3 Workflow

1. **Load** the raw acceleration data from the FEMTO-ST dataset.
2. **Extract features** (RMS, kurtosis, peak-to-peak, FFT band energies) for each snapshot.
3. **Standardize** the features (zero mean, unit variance) so that no single feature dominates.
4. **Train** Isolation Forest with `contamination=0.1`.
5. **Predict** a label for each snapshot: +1 (normal) or -1 (anomaly).
6. **Visualize** the anomalies over time against RMS and other features.

6 Results

6.1 Feature Evolution Over Time

Figure 4 shows how RMS, kurtosis and peak-to-peak evolve over the bearing's life.

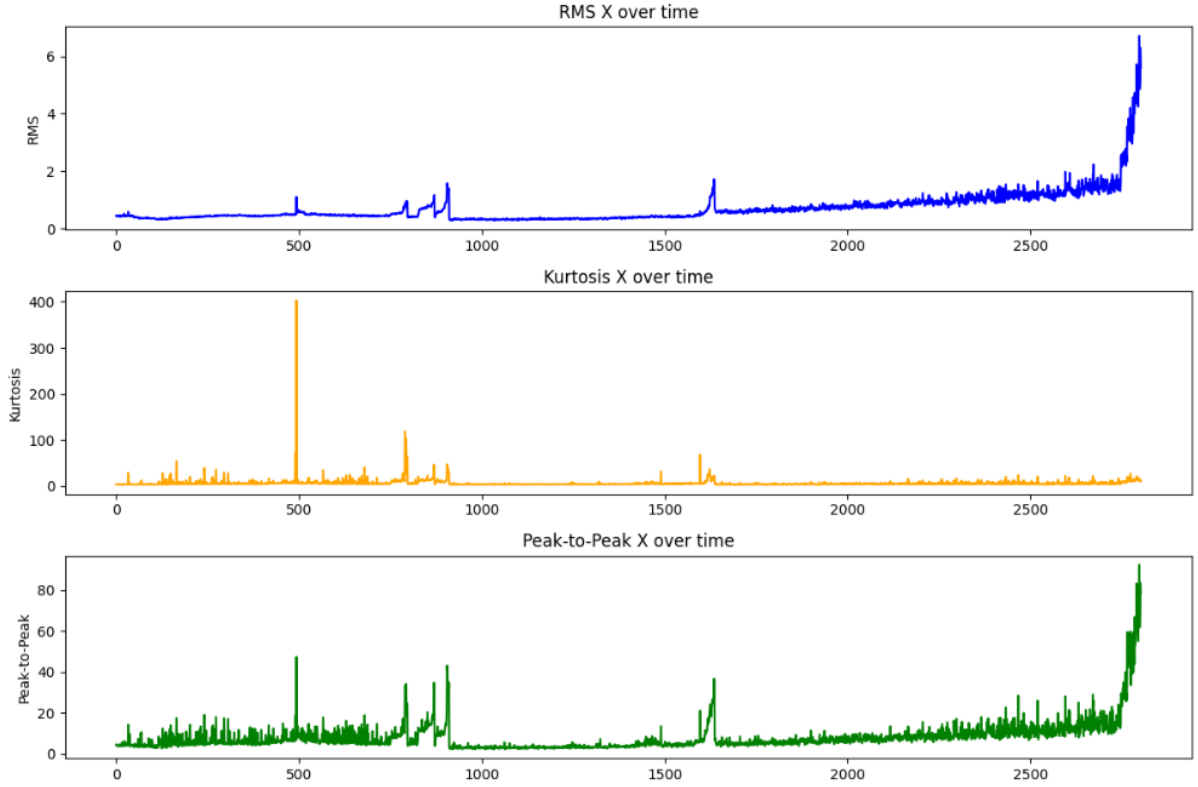


Figure 4: Evolution of time-domain features over the bearing's life. RMS increases gradually, kurtosis shows early spikes and peak-to-peak grows toward failure.

6.2 FFT Band Energy

Figure 5 shows the energy in the low, mid and high frequency bands.

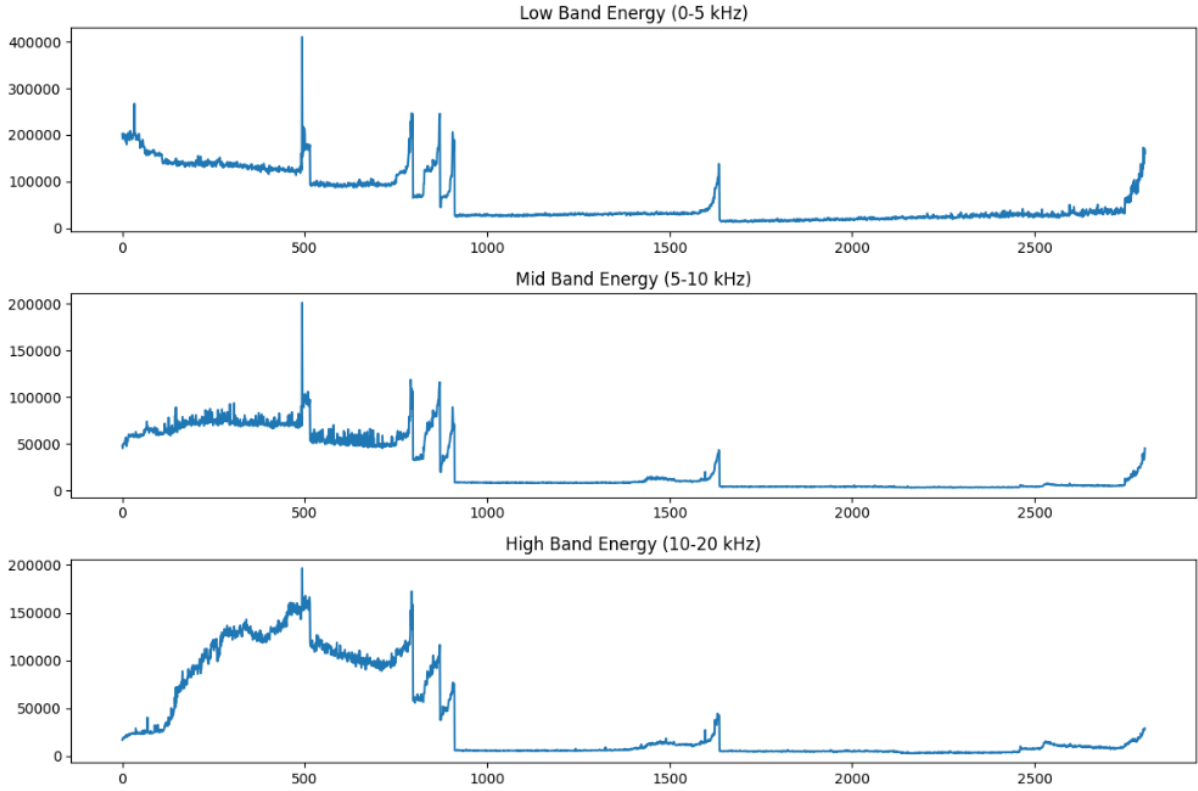


Figure 5: FFT band energies over time. High-frequency energy (10–20 kHz) rises first, providing an early warning of bearing degradation.

6.3 Anomaly Detection Results

Figure 6 shows the anomalies detected by the Isolation Forest, overlaid on the RMS curve.

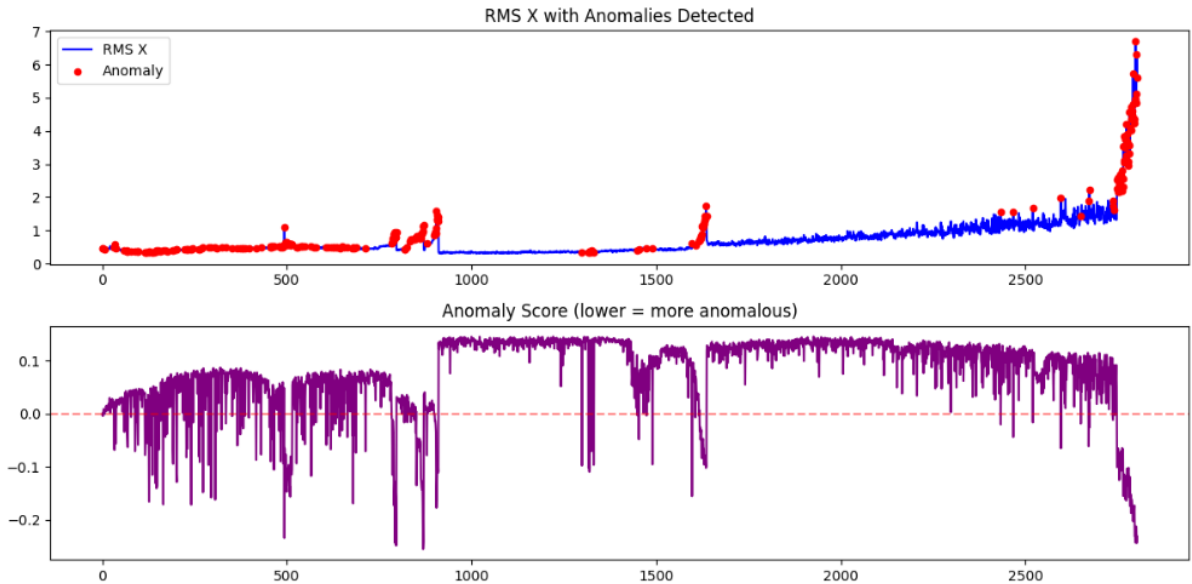


Figure 6: Anomalies detected by Isolation Forest. Red dots indicate anomalous snapshots. The model successfully identifies both early warning signs and final failure.

7 Discussion

7.1 What Worked

- Signal processing features (RMS, kurtosis, FFT bands) capture the degradation process.
- Unsupervised machine learning can detect faults without labeled data.
- Early fault signatures (kurtosis spikes, high-band FFT energy) appear before visible RMS changes.

7.2 Limitations

Important Limitations

- Dataset from a specific bearing type and operating condition; generalization is uncertain.
- **contamination** parameter (0.1) chosen heuristically.
- No ground-truth labels used to validate anomalies quantitatively.
- Operating condition changes also trigger anomalies; the model cannot distinguish a real fault from a benign change.

7.3 Lessons Learned

How Signal Processing Turns Data into Numbers

The raw sensor output is a long list of acceleration values. Features like RMS, kurtosis and FFT band energy are summaries of that list. They compress thousands of numbers into a few interpretable values describing the signal's energy, spikiness and frequency content.

How Anomaly Detection Works

Isolation Forest does not think. It learns the statistical distribution of normal data and flags points that are easy to isolate. The result is a label (+1 or -1) for each snapshot.

Anomaly Detection vs. RUL Prediction

Anomaly detection answers whether something is different right now. RUL prediction answers how much time is left. This project performs anomaly detection, the first step toward RUL prediction.

8 Conclusion

This project demonstrates a complete, reproducible pipeline for predictive maintenance using only free software and a public dataset. It combines signal processing (time-domain and frequency-domain features) with machine learning (Isolation Forest) to detect bearing degradation. The results show that early fault signatures can be detected automatically, enabling condition-based maintenance decisions.

9 Tools and Environment

Table 3: Tools used in this project.

Tool	Purpose
Google Colab	Cloud notebook environment (no installation)
Python 3	Programming language
pandas	Data loading and manipulation
NumPy	Numerical computation
scikit-learn	Isolation Forest, standardization
Matplotlib	Visualization
Hugging Face Datasets	FEMTO-ST dataset access
GitHub	Public portfolio hosting

10 References

1. FEMTO-ST / IEEE PHM 2012 Data Challenge, Bearing degradation dataset.
2. Hugging Face dataset: `Amgharr/FEMTO-ST_DATASET`.
3. F. T. Liu, K. M. Ting, Z.-H. Zhou, “Isolation Forest”, *IEEE ICDM*, 2008.
4. Nectoux et al., “PRONOSTIA: An Experimental Platform for Bearings Accelerated Degradation Tests”, *IEEE PHM*, 2012.

This project was developed as part of a self-directed research portfolio for the Mitacs Globalink Research Internship application.